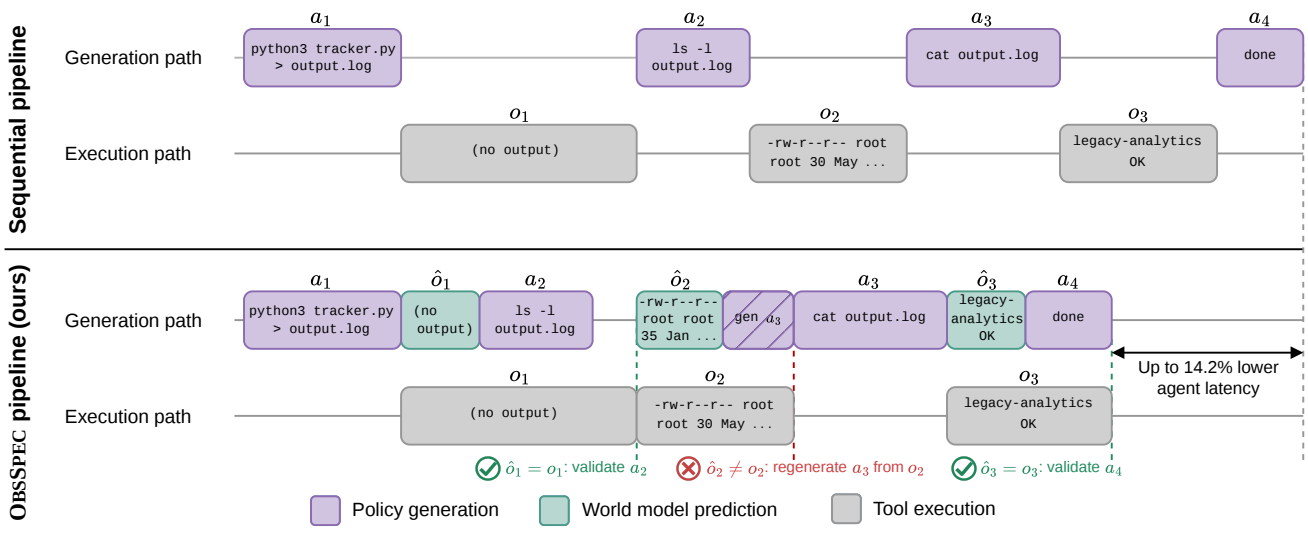

Task: run tracker.py with python3, write stdout to output.log, and verify the file contains exactly "legacy-analytics OK."
Sequential pipeline
Generation path
Execution path
$a_1$
python3 tracker.py > output.log
$a_2$
ls -l output.log
$a_3$
cat output.log
$a_4$
done
$o_1$
(no output)
$o_2$
-rw-r--r-- root root 30 May ...
$o_3$
legacy-analytics OK
OBSSPEC pipeline (ours)
Generation path
Execution path
$a_1$
python3 tracker.py > output.log
$\hat{o}_1$
(no output)
$a_2$
ls -l output.log
$\hat{o}_2$
-rw-r--r-- root root 35 Jan ...
gen $a_3$
$a_3$
cat output.log
$\hat{o}_3$
legacy-analytics OK
$a_4$
done
Up to 14.2% lower agent latency
$o_1$
(no output)
$o_2$
-rw-r--r-- root root 30 May ...
$o_3$
legacy-analytics OK
$\hat{o}_1 = o_1$: validate $a_2$
$\hat{o}_2 \neq o_2$: regenerate $a_3$ from $o_2$
$\hat{o}_3 = o_3$: validate $a_4$
Policy generation
World model prediction
Tool execution